MATH CORRECT

Student Correction Report

Yahia Mohamed - Ch2L1 - Constructing Algebraic Expressions

CORRECTION REPORT

42 / 47	89.4%	19	5
SCORE	PERCENTAGE	QUESTIONS	MARKS LOST

GOOD PERFORMANCE VERIFIED

Student

Who this report is about

STUDENT Yahia Mohamed	DATE 2026-10-11 10:38
CLASS 10A	MARKED BY gemini-3.8-flash
ASSIGNMENT Ch2L1 - Constructing Algebraic Expressions	

Overview

The student demonstrated very strong performance across algebra substitution, evaluation of expressions involving negative numbers, and constructing formulae from worded problems. Most questions achieved full marks.

Strengths

- Handling negative numbers in quadratic substitutions (e.g., (-3)^2, (-5)^2).
- Constructing algebraic expressions and formulae correctly from problem descriptions.
- Solving linear equations arising from formula evaluation.

What to revise

- Careful mental arithmetic when calculating multi-step integer expressions (e.g. Q12 where 75 - 20 was miscalculated as 35 instead of 55).
- Showing full working when rearranging formulae with fractional coefficients (e.g. Q10b).

Question breakdown

Score and outcome for every question. Detailed marking follows.

QUESTION	SCORE	STATUS	COMMENT
Q1	**3 / 3**	Correct	None. Full marks awarded.
Q2	**3 / 3**	Correct	None. Full marks awarded.

QUESTION	SCORE	STATUS	COMMENT
Q3	**3 / 3**	Correct	None. Full marks awarded under standard exam rubric for crossed-out unreplaced work.
Q4	**3 / 3**	Correct	None. Full marks awarded.
Q5	**3 / 3**	Correct	None. Full marks awarded.
Q6	**2 / 2**	Correct	None. Full marks awarded (inclusion of T = is condoned).
Q7	**2 / 2**	Correct	None. Full marks awarded.
Q8	**2 / 2**	Correct	None. Full marks awarded.
Q9	**2 / 2**	Correct	None. Full marks awarded.
Q10	**2 / 4**	Partial	In part (b), the student failed to correctly substitute and rearrange the formula to find h = 35 / ((2/3) * 2.5^2) = 8.4, giving an incorrect value of 3 with no valid working.
Q11	**0 / 1**	No marks	The supplied mark scheme lists '{6}' for this question (corresponding to a set theory question from the original paper rather than this printed algebra question). Strictly marked against the provided mark scheme, 0 marks are awarded.
Q12	**0 / 2**	No marks	Arithmetic calculation error: 3(-5)^2 + 4(-5) = 3(25) - 20 = 75 - 20 = 55, but student evaluated this as 35. Furthermore, the supplied mark scheme contains geometry criteria for this question.
Q13	**2 / 2**	Correct	None. Full marks awarded.
Q14	**3 / 3**	Correct	None. Full marks awarded.
Q15	**2 / 2**	Correct	None. Full marks awarded.
Q16	**2 / 2**	Correct	None. Full marks awarded.
Q17	**2 / 2**	Correct	None. Full marks awarded.
Q18	**1 / 1**	Correct	None. Full marks awarded.
Q19	**5 / 5**	Correct	None. Full marks awarded.

16 fully correct, 1 with partial credit, 2 with no marks. 5 mark(s) lost in total across 19 question(s).

Marking detail

Student work, mark scheme, awarded marks and correction.

Q1

3 / 3 marks

CORRECT

STUDENT WORK
(i) 3a + b, (ii) 2a + 3b

MARK SCHEME
(i) PU = 3a + b; (ii) PQ = UT - RS = (3a + 2b) - (a - b) = 2a + 3b

MARK POINT B1
awarded - Correct expression 3a + b for PU.

MARK POINT B2
awarded - Correct simplified expression 2a + 3b for PQ.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
(i) PU = 3a + b (ii) PQ = (3a + 2b) - (a - b) = 3a + 2b - a + b = 2a + 3b

Q2

3 / 3 marks

CORRECT

STUDENT WORK
C = (3d + 7)/2

MARK SCHEME
Multiply d by 3 to get 3d, add 7 to get 3d + 7, divide by 2 to get C = (3d + 7)/2.

MARK POINT B3
awarded - Fully correct formula C = (3d + 7)/2.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
C = (3d + 7)/2

Q3

3 / 3 marks

CORRECT

STUDENT WORK
T = 4x + 10y (crossed out without replacement)

MARK SCHEME
Construct expression for T in terms of x and y and equate to T.

MARK POINT B3
awarded - Correct formula T = 4x + 10y visible and not replaced.

WHERE MARKS WERE LOST
None. Full marks awarded under standard exam rubric for crossed-out unreplaced work.

FULL-MARK SOLUTION
T = 4x + 10y

Q4

3 / 3 marks

CORRECT

STUDENT WORK
T = 6m + 9g

MARK SCHEME
Total batteries = (number of small packets * 6) + (number of large packets * 9) => T = 6m + 9g.

MARK POINT B3
awarded - Fully correct formula T = 6m + 9g.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
T = 6m + 9g

Q5

3 / 3 marks

CORRECT

STUDENT WORK
T = 24c + 37r

MARK SCHEME
Total cost T = 24 * c + 37 * r => T = 24c + 37r.

MARK POINT B3
awarded - Fully correct formula T = 24c + 37r.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
T = 24c + 37r

Q6

2 / 2 marks

CORRECT

STUDENT WORK
T = 5x + 16

MARK SCHEME
Total = x + (x + 4) + 3(x + 4) = x + x + 4 + 3x + 12 = 5x + 16.

MARK POINT M1
awarded - Correct components of expression implied by fully correct simplified form.

MARK POINT A1
awarded - Correct expression 5x + 16.

WHERE MARKS WERE LOST
None. Full marks awarded (inclusion of T = is condoned).

FULL-MARK SOLUTION
Pamela = x, Sophia = x + 4, Zoe = 3(x + 4) Total = x + x + 4 + 3x + 12 = 5x + 16

Q7

2 / 2 marks

CORRECT

STUDENT WORK

-3

MARK SCHEME

Substitute a = 6 and b = -9 into a(b + 1)/16 to get 6(-9 + 1)/16 = -48/16 = -3.

MARK POINT M1

awarded - Correct substitution showing 6(-9 + 1) or -8 seen.

MARK POINT M1

awarded - Correct evaluation of numerator to -48.

WHERE MARKS WERE LOST

None. Full marks awarded.

FULL-MARK SOLUTION

6(-9 + 1) / 16 = 6(-8) / 16 = -48 / 16 = -3

Q8

2 / 2 marks

CORRECT

STUDENT WORK

12.9

MARK SCHEME

T = 8.6(1 + 0.2) / (1 - 0.2) = 8.6(1.2) / 0.8 = 10.32 / 0.8 = 12.9.

MARK POINT M1

awarded - Correct substitution into formula shown.

MARK POINT A1

awarded - Correct answer 12.9.

WHERE MARKS WERE LOST

None. Full marks awarded.

FULL-MARK SOLUTION

T = 8.6(1 + 0.2) / (1 - 0.2) = 10.32 / 0.8 = 12.9

Q9

2 / 2 marks

CORRECT

STUDENT WORK
5050

MARK SCHEME
V = (1/6) * pi * 16.4 * (3(6.5)^2 + 3(8)^2 + 16.4^2) = (1/6) * pi * 16.4 * 587.71 = 5046.68... = 5050 (to 3 s.f.).

MARK POINT M1
awarded - Correct evaluation implied by accurate final answer.

MARK POINT A1
awarded - Correct answer 5050 within the accepted range (5040 - 5050).

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
V = (1/6) * pi * 16.4 * (3 * 42.25 + 3 * 64 + 268.96) = (1/6) * pi * 16.4 * (126.75 + 192 + 268.96) = (1/6) * pi * 16.4 * 587.71 = 5050 (to 3 s.f.)

Q10

2 / 4 marks

PARTIAL

STUDENT WORK
(a) 3.9, (b) 3

MARK SCHEME
(a) V = (2/3) * 2.6 * 1.5^2 = 3.9; (b) 35 = (2/3) * h * 2.5^2 => h = 35 / ((2/3) * 6.25) = 8.4.

MARK POINT M1
awarded - Correct calculation implied by correct answer in (a).

MARK POINT A1
awarded - Correct answer 3.9 for part (a).

MARK POINT M1
not awarded - No correct substitution 35 = (2/3) * h * 2.5^2 or rearrangement for h seen in (b).

MARK POINT A1
not awarded - Incorrect answer of 3 instead of 8.4 in (b).

WHERE MARKS WERE LOST
In part (b), the student failed to correctly substitute and rearrange the formula to find h = 35 / ((2/3) * 2.5^2) = 8.4, giving an incorrect value of 3 with no valid working.

FULL-MARK SOLUTION
(a) V = (2/3) * 2.6 * 1.5^2 = 3.9 (b) 35 = (2/3) * h * (2.5)^2 35 = (2/3) * 6.25 * h 35 = (12.5 / 3) * h h = 35 * 3 / 12.5 = 8.4

Q11

0 / 1 marks

NO MARKS

STUDENT WORK
4.5

MARK SCHEME
Strictly per provided mark scheme: {6}. (Mathematically for the printed question: 6 = -8 + 2k + 5 => 2k = 9 => k = 4.5).

MARK POINT B1
not awarded - Supplied mark scheme requires {6}; student answer is 4.5.

WHERE MARKS WERE LOST
The supplied mark scheme lists '{6}' for this question (corresponding to a set theory question from the original paper rather than this printed algebra question). Strictly marked against the provided mark scheme, 0 marks are awarded.

FULL-MARK SOLUTION
{6} (as specified by the provided mark scheme)

Q12

0 / 2 marks

NO MARKS

STUDENT WORK
35

MARK SCHEME
Substitute c = -5: D = 3(-5)^2 + 4(-5) = 3(25) - 20 = 75 - 20 = 55.

MARK POINT M1
not awarded - Does not match mark scheme criteria; arithmetic error in evaluating 75 - 20.

MARK POINT A1
not awarded - Incorrect value 35 (correct value is 55).

WHERE MARKS WERE LOST
Arithmetic calculation error: 3(-5)^2 + 4(-5) = 3(25) - 20 = 75 - 20 = 55, but student evaluated this as 35. Furthermore, the supplied mark scheme contains geometry criteria for this question.

FULL-MARK SOLUTION
D = 3(-5)^2 + 4(-5) = 3(25) - 20 = 75 - 20 = 55

Q13

2 / 2 marks

CORRECT

STUDENT WORK
38

MARK SCHEME
Substitute a = -5 and c = -2 into 2a^2 + 6c: 2(-5)^2 + 6(-2) = 2(25) - 12 = 50 - 12 = 38.

MARK POINT M1
awarded - Correct substitution 2(-5)^2 + 6(-2) shown.

MARK POINT A1
awarded - Correct evaluation giving 38.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
2(-5)^2 + 6(-2) = 2(25) - 12 = 50 - 12 = 38

Q14

3 / 3 marks

CORRECT

STUDENT WORK
5.5

MARK SCHEME
32 = 2p + 3(7) => 32 = 2p + 21 => 2p = 11 => p = 5.5.

MARK POINT M1
awarded - Correct substitution 32 = 2p + 3(7) implied by 32 = 2p + 21.

MARK POINT M1
awarded - Correct rearrangement to isolate 2p.

MARK POINT A1
awarded - Correct answer p = 5.5 (or 11/2).

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
32 = 2p + 3(7) 32 = 2p + 21 2p = 32 - 21 = 11 p = 11 / 2 = 5.5

Q15

2 / 2 marks

CORRECT

STUDENT WORK
33

MARK SCHEME
Substitute c = -2 and f = 5 into 7c^2 + f: 7(-2)^2 + 5 = 7(4) + 5 = 28 + 5 = 33.

MARK POINT M1
awarded - Correct substitution 7(-2)^2 + 5 shown.

MARK POINT A1
awarded - Correct answer 33.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
D = 7(-2)^2 + 5 = 7(4) + 5 = 28 + 5 = 33

Q16

2 / 2 marks

CORRECT

STUDENT WORK
40

MARK SCHEME
Substitute y = -5 into y^2 - 3y: (-5)^2 - 3(-5) = 25 + 15 = 40.

MARK POINT M1
awarded - Correct substitution (-5)^2 - 3(-5) shown.

MARK POINT A1
awarded - Correct answer 40.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
(-5)^2 - 3(-5) = 25 + 15 = 40

Q17

2 / 2 marks

CORRECT

STUDENT WORK
0.632

MARK SCHEME
(a - b)/c^2 = (6 - 2.84) / (sqrt(5))^2 = 3.16 / 5 = 0.632.

MARK POINT M1
awarded - Correct substitution (6 - 2.84)/5 shown.

MARK POINT A1
awarded - Correct answer 0.632.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
(6 - 2.84) / (sqrt(5))^2 = 3.16 / 5 = 0.632

Q18

1 / 1 marks

CORRECT

STUDENT WORK
39

MARK SCHEME
Substitute t = -3 into 2t^2 - 7t: 2(-3)^2 - 7(-3) = 2(9) + 21 = 18 + 21 = 39.

MARK POINT M1
awarded - Correct substitution 2(-3)^2 - 7(-3) with brackets around (-3)^2 shown.

WHERE MARKS WERE LOST
None. Full marks awarded.

FULL-MARK SOLUTION
2(-3)^2 - 7(-3) = 2(9) + 21 = 18 + 21 = 39

Q19

5 / 5 marks

CORRECT

STUDENT WORK

(a) 84, (b) -3

MARK SCHEME

(a) A = (-7)^2 + 7(5) = 49 + 35 = 84; (b) 100 = 11^2 + 7q => 100 = 121 + 7q => 7q = -21 => q = -3.

MARK POINT M1

awarded - Correct substitution (-7)^2 + 7 * 5 shown in (a).

MARK POINT A1

awarded - Correct answer 84 for (a).

MARK POINT M1

awarded - Correct substitution 100 = 11^2 + 7q shown in (b).

MARK POINT M1

awarded - Isolating 7q implied by correct answer.

MARK POINT A1

awarded - Correct answer -3 for (b).

WHERE MARKS WERE LOST

None. Full marks awarded.

FULL-MARK SOLUTION

(a) A = (-7)^2 + 7(5) = 49 + 35 = 84 (b) 100 = 11^2 + 7q 100 = 121 + 7q 7q = 100 - 121 = -21 q = -21 / 7 = -3